The Anchor Accumulation Problem: Bounded Anchor Sets Recover Plasticity in Usage-Localized Elastic Weight Consolidation

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Abstract

Usage-localized elastic weight consolidation (DeLP) protects dendritic segments with per-neuron elastic anchors accumulated over the task sequence. A decomposition of average retention, $\bar{a}_T = \ell + \frac{T-1}{T}\,\mathrm{BWT}$, localized the cost of long task sequences not in the architecture but in the *learning term* ℓ: anchors accumulated under early tasks persist as an elastic drag on later learning. We call this the *anchor accumulation problem* and study it directly by bounding the anchor stack at M entries. Under four eviction criteria we find that (i) a FIFO budget recovers +4.5 and +9.0 points of final accuracy over unbounded DeLP at T=10 and T=20 tasks respectively (n=10 paired seeds, $t(9) > 19$ in both cases), recovering +5.4/+15.2 points of the learning term; (ii) the budget is a distinct control knob from the elastic weight λ: both trade protection for plasticity, but the budget controls the *age structure* of forgetting (which tasks are forgotten) while λ controls its *amount* (how much overall); (iii) eviction criteria keyed to an anchor's current constraint mass fail catastrophically — they preserve stale anchors created before the budget first binds ("grandfathering") and evict anchors at birth, losing −34.5 BWT

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— whereas age-first criteria are immune; trace instrumentation of anchor stacks shows the mechanism directly; and (iv) merging anchors removes exactly the redundant inter-anchor cross terms — the penalty drops one-sidedly by up to 64% while the accuracy trajectory remains indistinguishable from unbounded — separating the memory axis ($T \to M$ storage) from the plasticity axis. A budget of M=2 raises final accuracy to 68.2% at T=10, +13.0 points over unbounded DeLP, and the effect replicates on Fashion-MNIST at n=10 paired seeds (+13.6 points, $t(9)$=18.3) and on Split-CIFAR-100 with a ResNet-18 backbone, where at T=20 every bounded policy outperforms unbounded stacking (up to +5.0 points, $t(9)$=7.1) and anchor-stack traces replicate the age-versus-mass mechanism bit-for-bit. Single-anchor references bound the claim: plain EWC maximizes the learning term at 3–5× the forgetting, and online EWC with a tuned discount (γ<1) overtakes the budget at every grid cell — revealing that discounting is itself a soft bound, that soft and hard bounds fall on one curve relating the learning term to the protected-mass horizon (log-linear at the calibrated λ, monotone at every grid value), and that bounding accumulated anchor mass, not the eviction mechanism, is the active ingredient; on Split-CIFAR-100 a no-protection baseline remains above every arm. The explicit window's residual is an auditable, policy-selectable protection set; bounded anchor sets turn a diagnosed failure mode into a measured, tunable mechanism.

Keywords: continual learning, elastic weight consolidation, catastrophic forgetting, dendritic computation, stability–plasticity dilemma

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1. Introduction

Elastic weight consolidation (EWC) [1] and its variants protect important parameters by penalizing departure from values that served previous tasks. DeLP [2] localizes this protection to dendritic compartments and weights each neuron's penalty by its measured usage under previous tasks; an audit of the method established that usage weighting survives a parameter-budget cap (+1.79, p=0.005 at twenty tasks) while the full dendritic chain adds almost nothing once baselines share the hidden activation.

The same audit exposed a structural cost: DeLP appends one anchor set (posterior means and Fisher estimates) per task, so the elastic drag on optimization grows with the sequence length T. A decomposition of average retention,

$$\bar{a}_T \;=\; \ell \;+\; \frac{T-1}{T}\,\mathrm{BWT}, \tag{1}$$

separates the *learning term* ℓ (how well each task is acquired at its boundary) from the backward-transfer term. At ten tasks the losses of unbounded DeLP are concentrated in ℓ: anchors from early tasks persist as elastic drag long after the tasks they protect have been consolidated, and this is a property of the protection scheme, not of the architecture.

This paper treats the anchor stack itself as the object of study. We bound it at M entries and ask three questions: **(Q1)** Can the learning term be recovered by bounding, and at what stability cost? **(Q2)** Is the budget a redundant knob (a non-linear re-scaling of λ), or does it control something λ cannot? **(Q3)** If the stack must be bounded, *which* anchors should be evicted?

Contributions..

1. **Bounded anchor sets.** We introduce a family of eviction policies for anchor stacks and show that a FIFO budget recovers $+4.5/+9.0$ points of final accuracy at T=10/20 over unbounded DeLP (n=10 paired seeds, $t(9)$=21.4/19.1), by recovering $+5.4/+15.2$ points of the learning term (Eq. 1). A budget of M=2 reaches 68.2% at T=10 (+13.0 over unbounded), and the effect replicates on Fashion-MNIST (+13.6, n=10 paired seeds) and Split-CIFAR-100 (ResNet-18; at T=20 all six bounded policies beat unbounded, up to +5.0 points, $t(9)$=7.1).
2. **Two orthogonal knobs.** Through per-task accuracy profiles we show the budget controls the *age structure* of forgetting (which tasks degrade) while λ controls its total amount; the profiles are qualitatively different shapes, not re-scalings of one another (Fig. 2).
3. **A grandfathering failure mode.** Eviction criteria keyed to an anchor's current constraint mass systematically evict anchors *at birth* (mass ≈ 0 when the stack just changed) and preserve stale pre-budget anchors; BWT collapses to -34.5. Multiplicative age re-weighting does not fix this; an age-first, mass-second criterion does. Trace instrumentation of anchor stacks (Fig. 4) shows the mechanism directly.
4. **A compressed memory axis.** Fisher-cosine merging compresses the stack from T to M entries by removing exactly the cross terms between merged anchors: penalty instrumentation shows a one-sided drop (per-run maximum 18–64%) while the learned trajectory stays indistinguishable from unbounded, separating memory cost from plasticity.
5. **A mechanism-level boundary.** Sweeping online EWC's discount

$\gamma \in \{0.5, 0.9, 1\}$ shows a tuned discount overtakes the budget on accuracy at matched forgetting, and that soft (discount) and hard (window) bounds collapse onto one log-linear law between the learning term and the protected-mass horizon: bounding accumulated anchor mass is the active ingredient.

2. Related work

Regularization-based continual learning.. EWC [1] places a quadratic penalty around posterior estimates of previous tasks; SI [3] accumulates path-integral importances online; MAS [4] estimates sensitivity unsupervised. Anchor accumulation is not specific to DeLP: any scheme that stores per-task quadratics pays storage $O(T)$ and, if all terms persist at full weight, an increasingly rigid objective. Online EWC [5] addresses the storage axis by collapsing the stack into a single running posterior; with a discount $\gamma<1$ its Fisher mass is geometrically bounded, which makes it a *soft* bound on the same accumulated mass we diagnose (Sec. 5.1). Bounded anchor sets are complementary: they keep per-task terms and bound their number.

Parameter isolation and budgeting.. PackNet [6] and HAT [7] budget capacity explicitly. Our contribution differs: the budgeted object is the *protection* state, not the network capacity, and eviction can be nearly free when the penalty is combined correctly (merge).

Dendritic computation.. Active-dendrite networks [8, 9] provide the architectural substrate; DeLP's audit [2] established which part of the dendritic chain carries the benefit.

3. Background: DeLP and anchor accumulation

Setup.. A network is trained on a sequence of T tasks (permuted MNIST / Fashion-MNIST). DeLP freezes dendritic segments unused by the current task and adds, after each task t, an anchor set $\mathcal{A}_t = \{(\theta_t, F_t)\}$ (parameter snapshot and Fisher diagonal). The protection objective during task t is

$$\mathcal{L}_t(\theta) \;=\; \mathcal{L}_t^{\text{task}}(\theta) \;+\; \frac{\lambda}{2} \sum_{k<t} \sum_{i} F_i^{(k)} \, (\theta_i - \theta_i^{(k)})^2, \tag{2}$$

where the inner sum runs over the accumulated stack $\mathcal{A}_1, \ldots, \mathcal{A}_{t-1}$: the penalty is *unbounded* in T, both in storage and in constraint mass.

Where the cost lands.. Following [2], average retention decomposes as Eq. 1. At T=10 the learning term of unbounded DeLP sits 5.4 points below its T=5 value, and the gap widens with T; anchored identity Eq. 1 attributes the gap to ℓ, not BWT. We call this the *anchor accumulation problem.*

4. Bounded anchor sets

Budgeted stacks.. We replace the unbounded stack with a bounded one of size M. After appending the anchor for task t, if $|\mathcal{A}| > M$ an *eviction policy* removes or combines entries:

$$\mathcal{A} \leftarrow \mathsf{evict}_\pi(\mathcal{A} \cup \{(\theta_t, F_t)\},\, M)\,. \tag{3}$$

Policies.. Table 1 defines the six policies we compare. They span the design space along one axis: *what signal drives the choice* — age (fifo), chance (random), current constraint mass (redun), pairwise redundancy (merge), or combinations of age and mass (agew, agew2).

Table 1: Eviction policies for a stack over budget. Constraint mass $m_k = \sum_i F_i^{(k)}(\theta_i^{\text{now}} - \theta_i^{(k)})^2$ is measured at eviction time.

Policy	Key	Rule (when $\lvert\mathcal{A}\rvert > M$)
unbounded	—	never evict (original DeLP)
fifo	age	evict the oldest anchor
random	—	evict uniformly at random
redun	mass	evict $\arg\min_k m_k$
agew	mass × age	evict $\arg\min_k m_k \cdot (k{+}1)$, oldest $k{=}0$
agew2	age → mass	evict oldest anchor with below-median m_k; fallback fifo
merge	redundancy	merge the pair with most similar F (cosine): $F \leftarrow F_i + F_j$, $\theta \leftarrow$ weighted mean; repeat

Instrumentation.. Each anchor carries an *origin* tag (the task that created it), and we record the surviving-origin set after every phase (`survivor_history`). This turns the eviction dynamics into a directly observable object (Fig. 4).

5. Experiments

Protocol.. Permuted-MNIST and Fashion-MNIST, $T \in \{10, 20\}$; architecture, optimizer, permutation generation, and evaluation identical to the DeLP audit protocol [2]; λ=5000 unless stated. Every arm runs at the pipeline's *default* hidden-layer gain — the uncalibrated condition of the audit's calibration protocol [2] — and no arm is gain-tuned within this study; since the gain is shared, the confound the audit documented (a single gain

Table 2: Paired fifo–unbounded deltas, M=5, λ=5000, n=10 seeds.

Arm	Δacc	$t(9)$	ΔBWT
fifo T=10	$+4.51 \pm 0.67$	21.4	-1.50 ± 0.49
fifo T=20	$+8.98 \pm 1.49$	19.1	-7.20 ± 1.97

scalar moves accuracy by more than the entire method) cannot differentially favor any arm, only shift all arms together. Paired comparisons use identical seeds. Budgets $M \in \{2, 5\}$ anchor the main grid; a seven-point dose–response sweep $M \in \{2, 3, 4, 5, 6, 8, 10\}$ (T=20, λ=5000; Sec. 5.2) fills the interior, and no M was tuned post hoc. Section 5.6 states the Split-CIFAR-100 protocol separately (different backbone and protection pathway).

5.1. Bounding recovers plasticity (Q1)

Table 2: a FIFO budget at M=5 trades 1.5/7.2 points of BWT for +4.5/+9.0 accuracy. The mechanism is the learning term: ℓ rises by +5.4 and +15.2, exactly the recovery predicted if accumulated elastic drag were the binding constraint (Sec. 3); the grid in Table 3 shows the same signature for every bounded arm. Eq. 1 is an accounting identity, so the decomposition alone is not evidence; the causal claim rests on the eviction traces (Sec. 5.3) and on the λ-by-budget interaction (Sec. 5.2), which localize *when* the learning term moves — namely when anchors are actually retired. Corrected SI at T=10 (λ=100) reaches 40.3%.

Single-anchor references.. Because the diagnosis targets the *stack*, we also ran two protection schemes that cannot accumulate one, on the identical architecture, protocol, and λ grid (n=10 paired seeds): plain EWC, which keeps

Table 3: Policy grid, M=5, λ=5000, mean ± sd over n=3 seeds. ℓ = learning term (mean accuracy at task boundaries). agew is bit-identical to redun, as predicted in Sec. 5.3.

T	Arm	BWT	acc	ℓ
10	unbounded	-5.5 ± 0.6	55.3	60.2
10	fifo	-6.6 ± 0.9	59.7	65.6
10	random	-9.1 ± 2.6	60.2	68.4
10	redun	-18.3 ± 0.8	53.6	70.1
10	agew	-18.3 ± 0.8	53.6	70.1
10	agew2	-8.4 ± 0.9	60.8	68.3
10	merge	-5.5 ± 0.6	55.3	60.3
20	unbounded	-4.4 ± 1.0	33.7	37.9
20	fifo	-10.2 ± 1.9	43.4	53.1
20	random	-13.5 ± 0.4	42.0	54.8
20	redun	-30.2 ± 0.4	36.3	65.0
20	agew	-30.2 ± 0.4	36.3	65.0
20	agew2	-16.5 ± 0.3	42.3	58.0
20	merge	-4.5 ± 0.9	33.6	37.9

only the most recent task's Fisher and anchor (no accumulation, no running Fisher update — distinct from the original accumulated variant of [1]), and online EWC [5] with $\gamma \in \{0.5, 0.9, 1\}$, which collapses the stack into a single running posterior with geometrically discounted Fisher accumulation. All arms share the identical λ grid and are compared at each scheme's best grid value, so no scheme is disadvantaged by another's tuning. Plain EWC posts the highest final accuracy of any arm (68.3–71.1 at T=10, 56.4–59.4 at T=20,

best at λ=1000) by maximizing ℓ (81–85): with no stack there is no accumulation — and no retention either, at 3–5× the forgetting of unbounded DeLP (BWT $-15.8/-25.7$ at its best λ vs. $-5.5/-4.4$). Under the final-accuracy metric the learning-term dominance outweighs retention; a reader who weights BWT equally would rank it last. Online EWC at its best grid value (λ=1000) reaches 67.5/46.0: at T=10 it is statistically indistinguishable from DeLP-fifo at the same λ (67.4; paired $t(9)=-0.3$); at T=20 fifo is +3.5 ahead ($t(9)$=7.0). A collapsed posterior thus recovers most of the plasticity, but it posts neither the best long-sequence accuracy nor, by construction, any anchor accumulation problem to cure. Its forgetting is not unstructured: per-task forgetting (peak minus final accuracy; T=20, λ=1000, n=10) concentrates on older tasks for every arm, with drop–age correlations of −0.94 (plain), −0.79 (DeLP-fifo), −0.70 (online EWC), and −0.43 (unbounded). Sweeping the discount shows that the collapsed posterior does have an age knob after all: lowering γ to 0.9 and 0.5 raises ℓ from 51.1 to 56.6 and 76.1 (T=20, λ=1000), and at γ=0.5 online EWC overtakes DeLP-fifo at every grid cell we tested (paired +4.7 to +7.9, all $|t(9)| \geq 10.4$) at essentially matched BWT (−19.5 vs. −19.7). Discounting, however, is itself a bound: with geometrically decayed Fishers the accumulated mass converges to an effective horizon of $\approx 1/(1-\gamma)$ tasks, so γ<1 is a *soft* counterpart of the eviction window, and the two mechanisms turn out to be interchangeable points on one curve. Across all five bounded-mass arms at λ=1000, ℓ is log-linear in the protected horizon h (h=1 plain, h=2 for γ=0.5, h=5 for the window, h=10 for γ=0.9, h=20 unbounded): $\ell \approx 84 - 11 \ln h$ with R^2=0.99 — a fit that holds for every seed individually (per-seed $R^2 \geq 0.98$, slope -10.9 ± 0.6),

replicates at T=10 (R^2=1.00), and whose ordering is monotone at every grid λ (mean-fit R^2 0.99/0.94/0.80 as λ tightens). Bounding accumulated anchor mass — not the mechanism that enforces the bound — is the active ingredient, and the soft bound is the stronger remedy on final accuracy in this regime. What the explicit window retains is auditability and policy-level choice: its protection set is discrete and named by the survivor trace; the forgetting profile it produces is the one the trace predicts (8.6 on recent tasks vs. 27.7 on the evicted oldest); and the eviction key can be swapped (age, mass) or the stack merged without touching the update rule. The object studied here remains the stack; the budget is one of two cures, and on this benchmark the simpler one wins.

5.2. Budget vs. λ: two orthogonal knobs (Q2)

Fig. 1 shows the Pareto plane; Fig. 2 the per-task accuracy profiles; Table 4 the λ interaction.

Dose–response.. A seven-point sweep at T=20, λ=5000 (n=10) shows the budget's effect is monotone across the whole interior, not an artifact of two extreme points: final accuracy $52.6, 49.0, 44.7, 42.0, 40.1, 36.3, 35.1$ for M=$2, 3, 4, 5, 6, 8, 10$, with BWT recovering $-24.3 \rightarrow -5.2$ as the window widens toward M=10; every M remains significantly above unbounded (38.6; paired $t(9)$ from 27.2 at M=2 down to 8.0 at M=10, approaching the window's own limit).

The fifo advantage is robust to the choice of λ: even granting unbounded DeLP its best grid value at each T (λ=1000, the value at which unbounded is strongest at T=20), fifo retains a +10.9-point advantage there (and +3.5

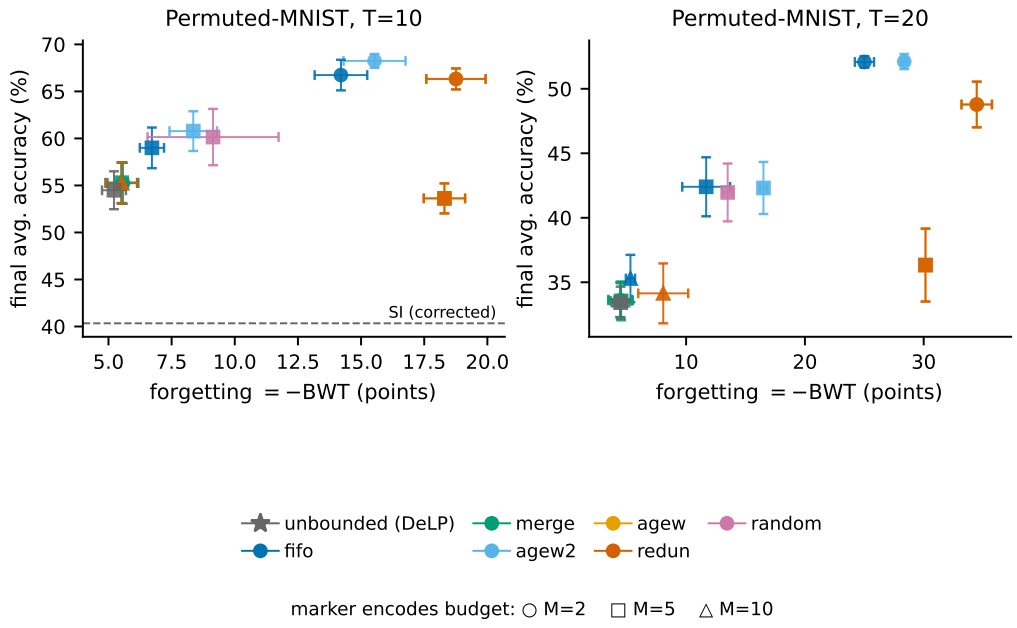


Figure 1: Accuracy–forgetting trade-off on Permuted-MNIST. Color encodes policy; marker encodes budget M. Error bars: sd over seeds (n=10 for unbounded/fifo M=5; else n=3). The smaller budget M=2 trades BWT for accuracy at both budgets in the same direction; merge coincides with unbounded at every M; redun sits dominated.

at T=10). That the unbounded-optimal λ must drift downward as T grows is itself a symptom of accumulating rigidity: the protection budget has to be relaxed sequence-wide to compensate for anchors that a bounded scheme would simply retire.

Both knobs move the system along a protection–plasticity spectrum, but they are not re-parameterizations of one another. At matched plasticity recovery, the two knobs produce *different age structures* of forgetting (Fig. 2): λ=1000 scales down protection uniformly — the profile keeps its steep age gradient (38.5-point spread, oldest tasks at 65–83%, newest at 45–51%) —

Table 4: λ interaction (M=5, n=10 paired seeds): the fifo advantage is positive at every λ and grows with T (all paired $t(9) \geq 11$, $p < 10^{-6}$).

λ	T	unbounded BWT/acc	fifo BWT/acc	Δacc
1000	10	−11.6/63.9	−10.8/67.4	+3.5
5000	10	−5.2/54.5	−6.7/59.0	+4.5
20000	10	−1.9/49.0	−4.0/53.1	+4.1
1000	20	−13.8/38.6	−19.7/49.5	+10.9
5000	20	−4.5/33.4	−11.7/42.4	+9.0
20000	20	−1.5/29.8	−6.6/37.3	+7.5

while fifo M=2 redistributes forgetting onto the oldest tasks, flattening the profile (25.7-point spread). No value of λ produces the flat profile, and no value of M produces the uniform scaling.

5.3. Which anchors to evict (Q3)

The grandfathering trap.. Mass-keyed eviction is not a straw man: "drop the anchor that constrains least" is the natural heuristic, homologous to low-sensitivity-first intuitions in pruning and in MAS-style importance estimates, and it is what a practitioner would try first. Its systematic failure is the point. Constraint mass m_k (Table 1) is measured at eviction time and is a function of staleness: anchors created before the budget first binds have $\theta^{(k)}$ far from θ^{now} and permanently large m_k; anchors created *after* the stack is full have $\theta^{(k)} \approx \theta^{\text{now}}$ and $m_k \approx 0$. Mass-keyed eviction therefore preserves exactly the stalest anchors and evicts new ones *at birth*: redun's stack freezes at $\{\mathcal{A}_1, \mathcal{A}_2\}$ (Fig. 4), the first two tasks keep strong protection (87/85% final accuracy at M=2), and tasks created later lose all protection within one phase (final

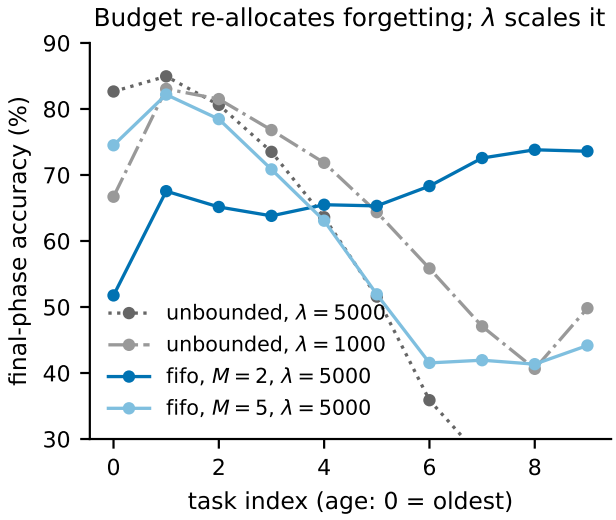


Figure 2: Final-phase accuracy per task (T=10, n=10). λ knobs (gray) translate the profile; budget knobs (blue) flatten it: forgetting is re-allocated to the oldest tasks.

accuracy 19–25%), driving BWT to -34.5 at T=20 despite the *highest* ℓ of any arm.

Re-weighting mass by age does not help.. agew (mass $\times$ age-rank) is *bit-identical* to redun across all seeds and configurations: mass differences span orders of magnitude and swamp the age factor, so the argmin never flips. Any criterion with mass as the primary key inherits the trap.

Age-first criteria close the matrix.. agew2 (oldest anchor with below-median mass; fallback fifo) behaves like fifo ($\pm$1.5 points): at M=2 its stack is $\{\mathcal{A}_1, \mathcal{A}_T\}$ — the oldest anchor's mass is always above median and survives, everything in between dies at birth. Table 5 summarizes the resulting 2$\times$2: the primary key, not the secondary signal, determines whether a criterion works.

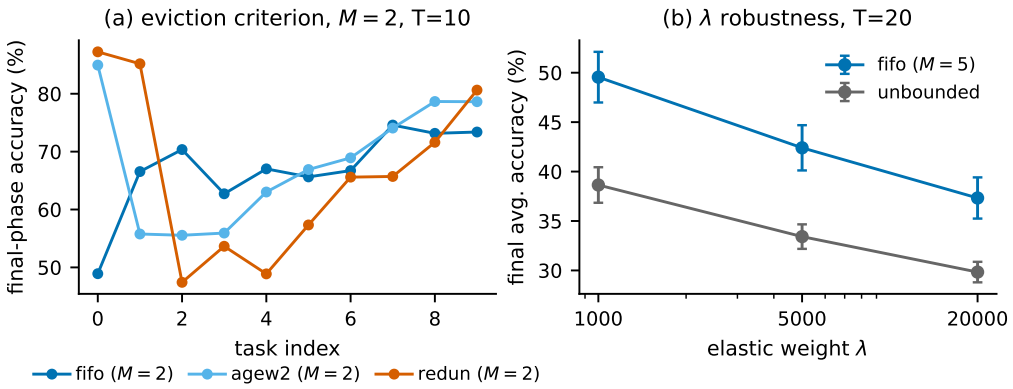


Figure 3: (a) Mass-keyed eviction (redun) grandfather-protects the first M tasks and erases the middle; age-first criteria (fifo, agew2) spread the cost. (b) The fifo advantage over unbounded holds at every λ.

5.4. Memory without behavior change: merge

Merging the most Fisher-similar pair removes exactly the cross term between the merged anchors. Penalty instrumentation over 40 runs ($T \in \{10, 20\}$, $M \in \{2, 5\}$, n=10; exactly $T-M$ merges per run) makes the change quantitative: the penalty decreases one-sidedly — never increases — with per-run maximum relative drops of 43–45% at M=2 (worst case 64%) and 18–34% at M=5, and 77% of individual merges move the penalty at all. The removed mass is precisely the redundancy the merge detected; the accuracy trajectory nonetheless remains indistinguishable from unbounded at every M (Table 3) while storing $M \ll T$ anchors (-75% at T=20, M=5). Merge and fifo are complementary: merge buys storage at a measured, behaviorally invisible penalty reduction but recovers no plasticity; fifo buys plasticity at a BWT price but no storage reduction.

Table 5: Criteria 2×2: primary key (rows) × secondary signal (columns). Only age-first criteria avoid the trap. T=10/T=20 BWT and accuracy, M=2 or M=5 as marked, n=3.

Primary key	Secondary	Outcome
mass	—	redun: grandfathering, BWT −30.2 (M5, T20)
mass	× age	agew ≡ redun (bit-identical)
age	—	fifo: sliding window, BWT −10.2 (M5, T20)
age	median mass	agew2 ≈ fifo (±1.5 pts)

5.5. Generality check: Fashion-MNIST

Fashion-MNIST, T=10, n=10 paired seeds: unbounded 35.1 ± 1.3; fifo M=5 38.1 ± 1.7 (+2.9, $t(9)$=11.5); fifo M=2 48.7 ± 2.5 (+13.6, $t(9)$=18.3). The budget effect replicates and amplifies as the budget shrinks.

5.6. Generalization to Split-CIFAR-100

Protocol.. Split-CIFAR-100, $T \in \{10, 20\}$ (10 classes per task), ResNet-18 adapted to 32×32 inputs with GroupNorm in place of BatchNorm — running statistics are buffers that drift under later tasks and silently destroy protected tasks while the anchored weights themselves are held fixed, so they are excluded by design. Evaluation and training are task-incremental with a masked softmax head: each task computes its loss only over its own 10 classes, so head rows of past tasks receive no gradient and are protected by construction; anchors therefore cover the backbone only, and forgetting can propagate exclusively through the shared trunk — exactly the pathway the anchor stack governs. Fisher diagonals are rescaled to unit global mean (one scalar over all parameters), making λ backbone-independent; gradients are clipped at norm 5. SGD, five epochs per task, λ=0.1 (chosen by an

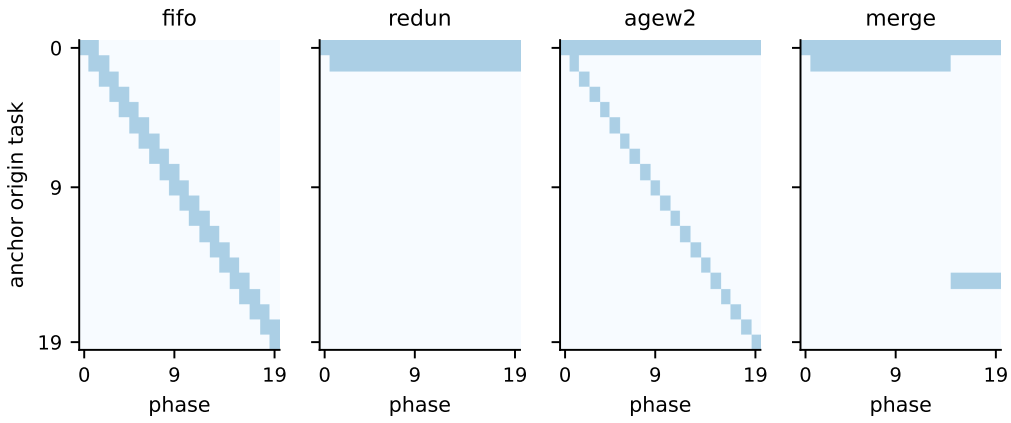


Figure 4: Anchor-stack composition over the task sequence (T=20, M=2, seed 0; identical for other seeds). Each column lists which anchors (by creation task) are alive at the end of each phase. fifo is a pure sliding window; redun freezes on the first two anchors (evicting every later anchor at birth); agew2 protects the oldest and the newest; merge partially grandfather-protects.

independent calibration sweep on unbounded DeLP only), n=10 seeds; eviction policies, origin instrumentation, and paired evaluation identical to the MNIST experiments. Unbounded DeLP never evicts, so its M is inert (final stack size T in every run); it is materialized once per T and serves as the reference for both budgets. All CIFAR comparisons are task-incremental (masked head); absolute accuracies are therefore not comparable to the class-incremental Split-CIFAR-100 literature, and every claim in this section is a within-family, same-protocol paired comparison.

The budget effect strengthens with T on natural images.. At T=20 every bounded policy outperforms unbounded stacking (Table 6): fifo +5.0 ($t(9)$=7.1), agew +4.2 (t=7.3), random +3.6 (t=9.2), redun +3.4 (t=4.0); at T=10 the effect is marginal, about +1.5 points at $t(9) \approx 2$. The dose–

Table 6: Split-CIFAR-100, λ=0.1: final accuracy, mean ± sd over n=10 seeds; parenthesized values are the paired difference Δacc against unbounded DeLP at the same seed (quoted in text with paired $t(9)$ where significant).

	T=10		T=20	
Policy	M=2	M=5	M=2	M=5
unbounded	23.3 ± 2.6		29.8 ± 4.2	
fifo	24.8 ± 2.8 (+1.5)	23.5 ± 2.6 (+0.2)	$\mathbf{34.8 \pm 4.8}$ (+5.0)	32.2 ± 4.3 (+2.4)
random	24.7 ± 2.4 (+1.5)	22.4 ± 2.7 (−0.9)	33.4 ± 3.9 (+3.6)	32.2 ± 4.7 (+2.4)
redun	24.0 ± 2.5 (+0.7)	23.6 ± 2.0 (+0.3)	33.2 ± 4.7 (+3.4)	32.9 ± 3.3 (+3.1)
merge	22.3 ± 1.7 (−1.0)	21.8 ± 3.2 (−1.5)	30.8 ± 5.2 (+1.0)	29.3 ± 3.4 (−0.5)
agew	24.0 ± 3.8 (+0.7)	22.7 ± 3.4 (−0.6)	34.0 ± 4.3 (+4.2)	31.6 ± 3.7 (+1.8)
agew2	23.7 ± 2.6 (+0.4)	22.3 ± 3.3 (−1.0)	32.5 ± 3.7 (+2.7)	31.1 ± 3.4 (+1.3)

response matches the mechanism: anchor drag accumulates with sequence length, so the benefit of bounding it should grow with T. As in Sec. 5.1, the gain lands in the learning term: fifo M=2 raises ℓ from 31.4 to 36.8 while BWT moves only from −1.7 to −2.1 — the budget buys plasticity, the trade Eq. 1 predicts. The decomposition itself is exact on these runs: over all 260 CIFAR records, $|\bar{a}_T - (\ell + \frac{T-1}{T}\mathrm{BWT})|$ never exceeds 5×10^{-3}, i.e. the rounding of two-decimal reporting.

A no-protection reference.. λ=0 (plain SGD, same protocol, n=10 paired seeds) reorders the picture and must be reported: 27.0 ± 1.7 at T=10 and 36.9 ± 3.4 at T=20 — above every DeLP arm, including fifo M=2 (+2.1, $t(9)$=1.5, not significant). The decomposition explains why: protection roughly halves the learning term (ℓ is 25.7 and 31.4 at T=10/20 for unbounded DeLP, vs. 46.1 and 64.3 at λ=0), while in this regime retention is

nearly free to forgo — even the large BWT of the unprotected arm ($-21.3/-28.9$) costs less than the ℓ it gains. Under task-incremental evaluation with a fixed masked head, trunk features are shared across tasks and trunk drift is cheap; regularization-based protection at any λ is then a net tax on accuracy. The budget effect remains real within the protected family, but on this benchmark bounding is necessary, not sufficient: protection itself must be checked against a no-protection reference before the mechanism is credited.

Stack traces replicate bit-for-bit.. At T=20, M=2, all ten seeds agree: fifo's stack is the sliding window $\{\mathcal{A}_{T-1}, \mathcal{A}_T\}$; redun freezes on $\{\mathcal{A}_1, \mathcal{A}_2\}$ (grandfathering); agew inherits the same freeze (eviction-time mass swamps the age factor); agew2 holds $\{\mathcal{A}_1, \mathcal{A}_T\}$ in 6/10 seeds (the age-primary hybrid); merge grandfather-protects partially. The mechanism isolated on Permuted-MNIST (Fig. 4) is not an artifact of that domain (Fig. 5).

A boundary condition on the mass-keyed failure.. Unlike on MNIST, the *secondary* effect — *which* criterion evicts — is weak here: fifo − redun is +1.6 ($t(9)$=2.0) at T=20, M=2, and agew2 − redun is slightly negative. The reason is structural: under masked-CE training, forgetting propagates only through the shared trunk, whose low layers are described similarly by every task's Fisher diagonal; even the stale anchors $\{\mathcal{A}_1, \mathcal{A}_2\}$ constrain features later tasks still use, so grandfathering them costs little. Mass-keyed eviction fails when anchors describe task-specific structure that goes stale; the primary effect of bounding (recovering plasticity) survives, while the relative cost of grandfathering scales with how much low-level structure tasks share — a testable prediction for domain-incremental sequences, where the gap should re-open.

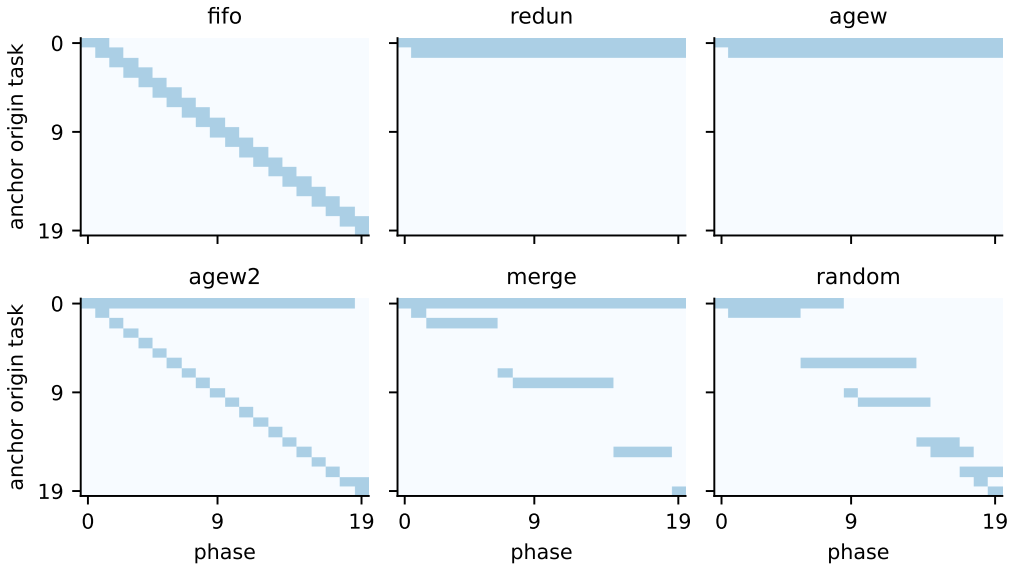


Figure 5: Anchor-stack composition on Split-CIFAR-100 (T=20, M=2, seed 0; all seeds agree on the final sets). Same rendering as Fig. 4. fifo is a sliding window; redun and agew freeze on the first anchors (grandfathering); agew2 pairs the oldest with the newest; merge partially grandfather-protects; random walks down.

6. Discussion

Why bounding works.. Eq. 1 says total retention is learning plus transfer; unbounded stacks tax learning with stale constraints. A FIFO window keeps only the freshest M constraints, whose $\theta^{(k)}$ are near θ^{now} and whose Fisher mass is relevant to the current basin: the penalty constrains what actually needs constraining.

The mass signal is not useless — it is mis-keyed.. redun asks "which anchor constrains least *now*?"; the honest answer at eviction time is "the one just created", which is exactly the wrong anchor to drop. agew2 shows mass adds marginal, unstable value as a secondary signal. The information structure

of the problem — eviction-time mass confounds relevance with staleness — makes age the only reliable primary key.

Limitations.. (1) Task-incremental sequences (permuted MNIST, Fashion-MNIST, Split-CIFAR-100 with a masked head); domain-incremental settings, where tasks share nearly all low-level structure, may shrink the age-over-mass gap further (Sec. 5.6). (2) $T \leq 20$; the ℓ-recovery trend grows with T but the BWT cost does too. (3) All MNIST arms share an uncalibrated hidden gain, as in the audit; the CIFAR study uses its own protocol with λ calibrated on unbounded DeLP only; conclusions are within-family comparisons. (4) Merge's penalty change is measured by instrumentation (Sec. 5.4), not bounded formally; a closed-form error bound is future work. (5) On Split-CIFAR-100 the no-protection reference outperforms every protection arm (Sec. 5.6): task-incremental evaluation makes forgetting cheap, so the anchor pathway is a net tax there and the budget result is internal to protected arms.

7. Conclusion

Bounding the anchor stack converts the diagnosed anchor accumulation problem into a control knob: FIFO eviction recovers the learning term at a declared BWT cost, the budget's effect is structurally different from λ's, age is the only sound eviction signal, and merge decouples memory from behavior. The effects replicate across three benchmarks — permuted and class-incremental, MLP- and ResNet-based — with the age-over-mass advantage bounded by a stated boundary condition on task relatedness and the protection pathway itself bounded by a no-protection reference on Split-CIFAR-100.

Single-anchor schemes cannot accumulate the problem by construction, and a discounted posterior — itself a soft bound, its mass converging to a horizon of $\approx 1/(1-\gamma)$ tasks — recovers plasticity through the same law: across soft and hard bounds the learning term falls log-linearly with the protected-mass horizon at the calibrated λ (monotone at every grid value), and a tuned discount overtakes the window on accuracy at matched forgetting. Bounding accumulated anchor mass, not the eviction mechanism, is the active ingredient; the explicit window is the variant whose protection set is auditable and whose policy is a choice. Protection schemes for continual learning should ship with bounded — or at least mergeable — anchor state whenever protection is warranted at all; the Split-CIFAR-100 reference (Sec. 5.6) shows that the warrant itself is regime-dependent and must be checked first.

Appendix A. Hyperparameters and reproducibility

Permuted-MNIST / Fashion-MNIST (Secs. 5 to 5.5).. Architecture as in the audit [2]: a 784–256 fully connected trunk whose pre-activation is modulated by $K{=}T$ dendritic segments (linear context-to-segment map, winning-segment readout), hidden activity sparsified by k-WTA with $k{=}64 = N/4$, linear 10-way head. Adam, lr 10^{-3}, batch 256, two epochs and 10,000 samples per task; permutation matrices drawn once per seed and shared by all arms. Diagonal Fisher and usage statistics from 300 and 500 samples per task respectively (max-log-probability Fisher variant); $\lambda_{DeLP}{=}5000$, corrected-SI reference $\lambda{=}100$. Paired arms use seeds 0–9; the policy grid seeds 0–2 (Table 3); λ grid as in Table 4. Every run seeds `torch.manual_seed` before model construction. Code and per-seed JSON records (including

`survivor_history` and `acc_matrix`) are released with the repository.

Split-CIFAR-100 (Sec. 5.6).. ResNet-18 adapted for 32×32 inputs (3×3 stem, stride 1, no max-pool) with all BatchNorm layers replaced by GroupNorm (32 groups). SGD, momentum 0.9, weight decay 5×10^{-4}, lr 0.05, batch 128, five epochs per task. Masked cross-entropy over the task's 10 classes; task-incremental evaluation over the same classes. EWC penalty over backbone parameters only, λ=0.1; diagonal Fisher estimated from 50 single-sample draws of the squared gradient of the max log-softmax on the finished task, rescaled to unit global mean; gradient clipping at max norm 5. Tasks are contiguous 10-class blocks of the standard class order; seeds 0–9 control init, shuffle, and eviction randomness. Per-job crash-safe checkpoints and the full JSON records (including `survivor_history` and `acc_matrix`) are released with the code.

Appendix B. Full per-task accuracy matrices

Final-phase accuracy (%) per task, Permuted-MNIST, λ=5000, M=5: mean over seeds (n=10 for unbounded and fifo, n=3 for random/redun/merge). These underlie Fig. 3a and the grandfathering analysis of Sec. 5.3.

CRediT authorship contribution statement

Ranxi Zhou: Conceptualization, Methodology, Software, Investigation, Formal analysis, Data curation, Writing — original draft, Writing — review & editing, Visualization.

Table B.7: T=10.

task	unbounded	fifo	random	redun	merge
1	82.6	74.5	83.0	83.5	82.3
2	84.9	82.2	82.9	85.5	84.6
3	80.6	78.5	79.3	82.1	81.7
4	73.5	70.8	71.9	73.3	72.6
5	63.6	63.1	62.5	66.3	65.7
6	51.6	51.9	38.0	18.7	52.7
7	35.9	41.5	49.2	22.4	36.0
8	27.0	41.9	39.9	20.8	29.3
9	23.0	41.3	42.3	25.4	24.9
10	22.1	44.2	52.3	58.2	23.1

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All code, experiment scripts, and the complete JSON records of every run (including per-phase accuracy matrices and anchor-stack survivor histories) are publicly available at `https://github.com/zhouranxi/delp-anchor-budget`. A preprint of the precursor audit study is archived on Zenodo (doi:10.5281/zenodo.23602281, version 2).

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Table B.8: T=20.

task	unbounded	fifo	random	redun	merge
1	78.6	59.3	70.7	84.4	80.6
2	84.2	73.2	78.6	83.4	83.8
3	79.6	68.8	69.6	78.3	79.7
4	71.7	58.1	56.4	69.3	70.5
5	61.3	48.4	48.4	61.1	60.6
6	50.8	40.3	46.1	11.6	51.8
7	37.9	37.0	38.1	16.3	36.6
8	28.9	38.3	31.7	16.9	30.2
9	23.4	34.0	36.7	17.4	26.2
10	18.6	34.4	35.4	13.5	17.4
11	14.5	34.0	21.4	14.8	14.4
12	13.7	36.3	30.7	14.9	11.0
13	14.5	35.1	28.1	19.4	14.7
14	12.9	34.6	28.8	20.2	11.6
15	13.5	34.9	30.0	25.1	16.2
16	14.8	36.5	42.3	30.9	18.5
17	12.0	36.3	38.0	23.7	13.9
18	12.1	36.3	31.2	30.4	9.7
19	12.1	33.2	36.0	35.0	9.3
20	13.3	39.2	41.2	60.1	14.4